PampelMuse

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Abstract

PampelMuse is a software package with the purpose of facilitating the analysis of integral-field spectroscopic observations of crowded stellar fields. It provides several subroutines to perform the individual steps of the data analysis.

1 Getting started

1.1 Requirements

PampelMuse is written in *python*. A recent version of *python* should be available on the system, *python v2.7* is recommended. In addition, several *python* packages are required to run PampelMuse:

- *numpy*[1], tested for version 1.6.1
- *scipy*[2], tested for versions ≥0.9, a version ≥0.10 is recommended because a faster algorithm for sparse matrix inversion was introduced in v0.10
- *astropy*[3], tested for version 0.2.5

Note that the versions of the packages listed above are just a recommendation. The code may also run with earlier versions. In addition to these packages, it is highly recommended that the following packages are available on the system:

- *matplotlib*[4], tested for version 1.1.1
- *PyQt4*[5]

These packages are required in order to work with the graphical user interface (GUI) that is provided with PampelMuse. When starting PampelMuse the availability of *matplotlib* and *PyQt4* is checked. If the packages are not available, the subroutines that implement the GUI will be disabled. Finally, the packages listed below are not required to run the core packages of PampelMuse, but some auxiliary routines depend on them:

- *astLib*[6], tested for version 0.7.0

[1] `http://www.numpy.org/`
[2] `http://scipy.org/`
[3] `http://www.astropy.org/`
[4] `http://matplotlib.org/`
[5] `http://www.riverbankcomputing.co.uk/software/pyqt/intro`
[6] `http://astlib.sourceforge.net/`

- *pyraf*[7]

1.2 Download and Installation

There are currently two possibilities to obtain PampelMuse that will be described in the following.

1.2.1 Using the tarball

A zipped version of the code is available at `http://path.to/pampelmuse.tar.gz`. After the download is completed, change to the folder containing the tarball and execute the following commands

```
# tar -xvzf pampelmuse.tar.gz
# cd pampelmuse/
# python setup.py install
```

This will install PampelMuse to your default *python* path. Note that the last step usually requires root privileges. If you do not have those privileges, you can define a custom path using the `--prefix` or `--home` arguments. You only have to make sure then that the path you are using to install PampelMuse is included in your `PYTHONPATH`. For example, the following commands will install PampelMuse to folders `$HOME/lib` and `$HOME/bin`.

```
# python setup.py install --home=~
# export PYTHONPATH=$PYTHONPATH:$HOME/lib/python ## for a bash-shell or
# setenv PYTHONPATH $PYTHONPATH:$HOME/lib/python ## for a c-shell
```

You probably want to store the modification of your `PYTHONPATH` variable in your shell configuration file (`.bashrc` or `.cshrc`), otherwise you have to set the variable again every time you open a new shell. For further informations on custom installation paths, check the *python* documentation.[8]

Afterwards, you should be able to execute

```
# PampelMuse --help
```

If no error message is returned, you have installed `PampelMuse` successfully!

1.3 Using the SVN repository

The maintenance of `PampelMuse` is handled via an SVN repository located at `https://subversion.assembla.com/svn/kamann-code.2`. The advantage of using the repository is that it allows you to always use the most recent version of the code. However, the disadvantage is that when updating the code I sometimes commit revisions that are not fully operational. So when updating to a new revision, be sure to check the most recent log entries.

To install from the SVN repository, first cd into the folder were you want to install the code and check out the repository:

```
# svn checkout https://subversion.assembla.com/svn/kamann-code.2/trunk
```

This should download the most recent version of `PampelMuse`. To use the code afterwards, the following modifications should be added to your shell configuration:

```
# ## for a bash shell:
# export pampelmuse_path=<path to PampelMuse> ## folder were SVN checkout was
```

[7] `http://www.stsci.edu/institute/software_hardware/pyraf`
[8] `http://docs.python.org/2/install/#alternate-installation`

```
performed
       # export PYTHONPATH=$PYTHONPATH:$pampelmuse_path
       # export PATH=$PATH:$pampelmuse_path/pampelmuse/bin
```